FIELD TRACE

AI markup assist for Bluebeam Revu 21 — sample output set

This set shows what Field Trace would produce from a batch of piping photos. Each sheet pairs the original field photo, carrying **editable markups** (standard PDF annotations: polylines, callouts, dimensions, revision clouds and a stamp, on named layers), with an **as-built isometric** drafted from the same traced geometry. The last page rolls the accepted markups up into a quantity takeoff.

The markups are a first pass. They are meant to be reviewed, corrected and signed off in Bluebeam Revu 21.11 by the person who does this work today. Anything under 70% confidence is dashed and flagged for verification. A photo that is too uncertain is held instead of auto-drafted (see FT-06).

Bluebeam OCR can make text in a photo searchable, such as stencils, tags and stamps. Field Trace works from the pipe geometry itself: it traces runs, places symbols and dimensions, and drafts the drawing. Field Trace is a Mimetic Technology prototype. It is not a Bluebeam product.

PHOTO CREDITS

FT-01	Shell and Tube Heat Exchanger Piping 1	Ivangiesen	CC0 1.0
FT-02	Ventilation piping installed in the back of house area of the future LIRR passenger concourse (CM014B, 03-19-2019)	MTA Capital Construction Mega Projects	CC BY 2.0
FT-03	Insulating piping in the back of house area of the future LIRR passenger concourse (CM014B, 03-27-2019)	MTA Capital Construction Mega Projects	CC BY 2.0
FT-04	Installing copper piping that will connect to a heating and cooling unit in the LIRR passenger concourse (CM014B, 1-24-2019)	MTA Capital Construction Mega Projects	CC BY 2.0
FT-05	Mechanical room.jpg	P199 at English Wikipedia	CC BY 2.5
FT-06	Installing hangers for piping along the east service corridor of the future LIRR concourse (CM014B, 08-30-2018)	MTA Capital Construction Mega Projects	CC BY 2.0

All photos are from Wikimedia Commons. They are not Schilke job photos. Project names, dates, tags and confidence values are sample data.

SHEET INDEX

PG	SHEET	TITLE	LOCATION	MARKUPS	ISSUES	STATUS
1	COVER	Cover, index, credits				
2	FT-01	Plate HX-1 — tower / condenser water	Mech. Rm B-104 · North wall	30	2	Draft — for review
3	FT-02	HWS offset at CMU wall — rough-in	Level B1 · Corridor C, column line 4	23	2	Draft — for review
4	FT-03	Overhead mains — east service corridor	Level B1 · East service corridor	20	2	Draft — for review
5	FT-04	Domestic water drops at wall penetrations	Level 1 · Back-of-house wall W-12	28	3	Draft — for review
6	FT-05	LP steam header & pump set	Central plant · Pump room	20	1	Reviewed
7	FT-06	Ceiling space above corridor — low light	Level B1 · Corridor D ceiling	14	2	Needs attention
8	TAKEOFF	Quantity takeoff — all photos				

MARKUP LAYERS

Pipe Runs	Polyline along each pipe centerline, colored by system
Fittings	Elbows, tees, couplings, flanges, caps
Valves & Devices	Valves, instruments, equipment callouts
Hangers & Supports	Hangers, strut, penetrations / firestop
Dimensions	Dimensions with basis noted (CMU coursing, pipe OD)
Issues	Revision clouds with comments (punch / QA)
Labels	System + size tags, pipe markers read, stamp

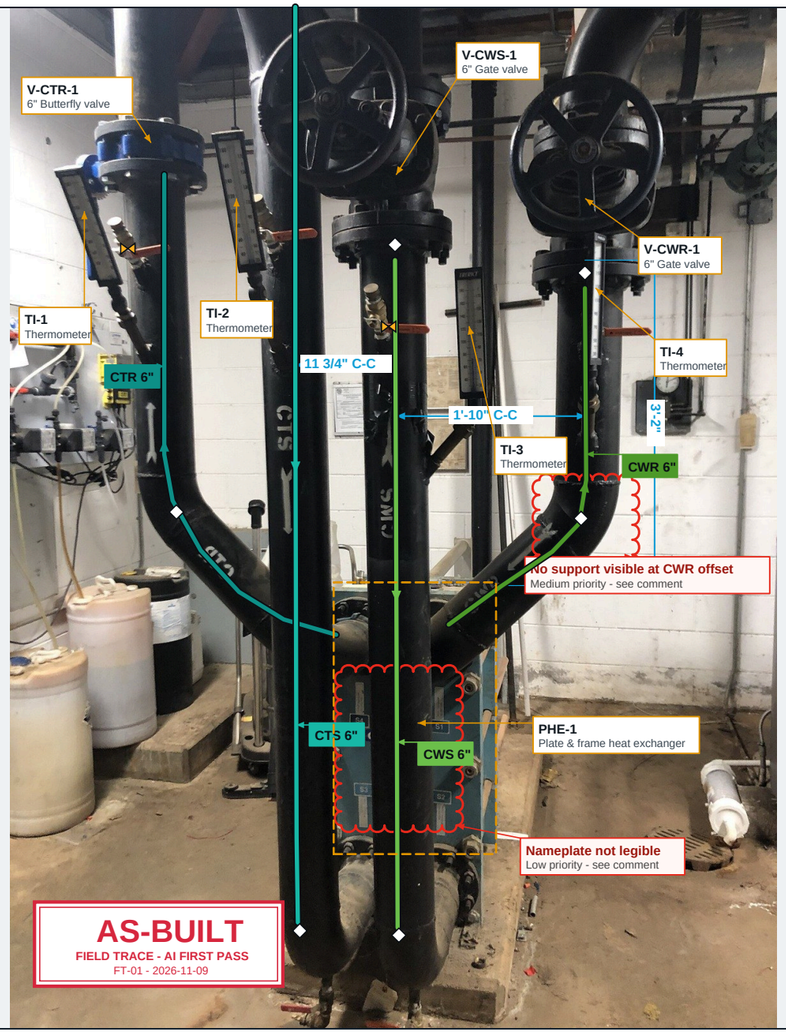
SYSTEM LEGEND

CTS	Cooling Tower Supply
CTR	Cooling Tower Return
CWS	Condenser Water Supply
CWR	Condenser Water Return
CHWS	Chilled Water Supply
CHWR	Chilled Water Return
HWS	Heating Hot Water Supply
HWR	Heating Hot Water Return
DCW	Domestic Cold Water
DHW	Domestic Hot Water
LPS	Low Pressure Steam
FP	Fire Protection
UNK	Unidentified — verify

FIELD TRACE

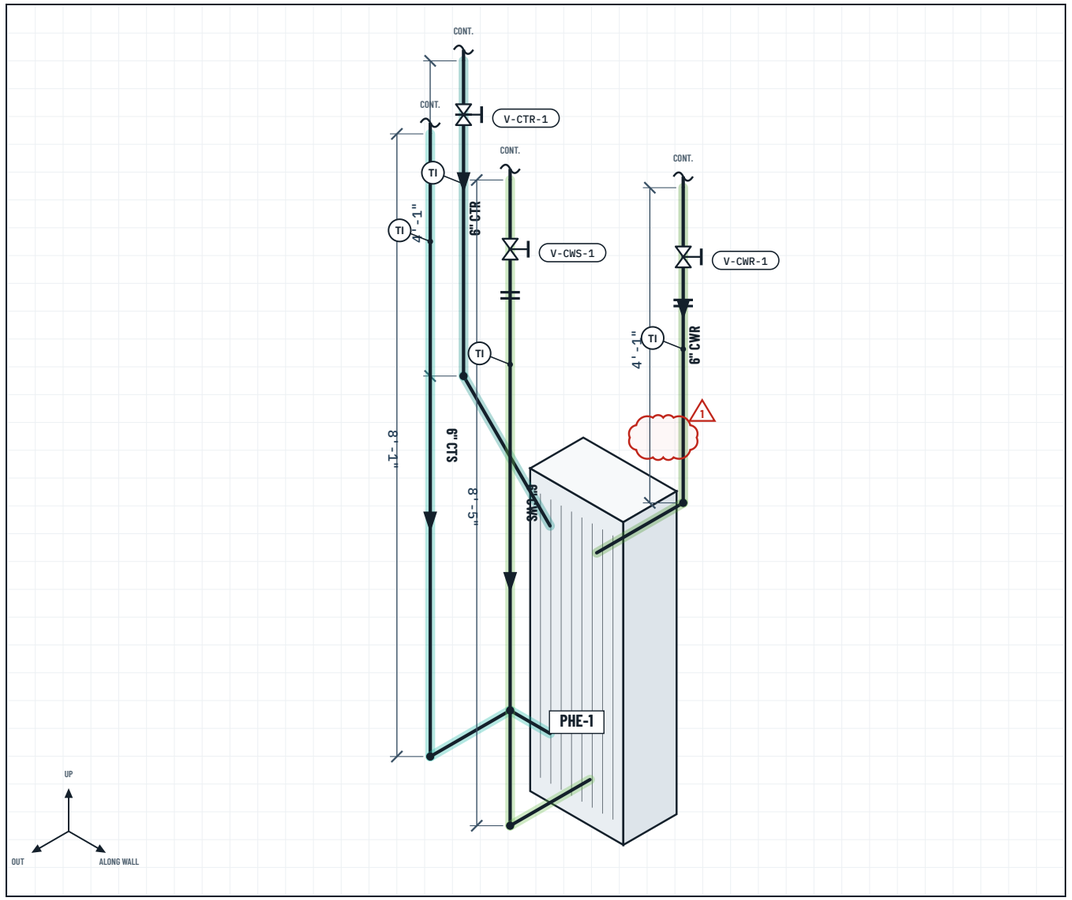
ANNOTATED FIELD PHOTO + AS-BUILT ISOMETRIC · FT-01

MIMETIC TECHNOLOGY PROTOTYPE · PREPARED FOR SCHILKE CONSTRUCTION · **SAMPLE DATA — NOT FOR CONSTRUCTION**



FIELD PHOTO FT-01 · MECH. RM B-104 · NORTH WALL · 2026-11-09 07:42

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — PLATE HX-1 — TOWER / CONDENSER WATER

N.T.S. · DERIVED FROM PHOTO FT-01 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 30 MARKUPS · 2 ISSUES · 0 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	6" CTR	Pipe Runs	None	95%
Pipe Tag	CTR 6"	Labels	None	95%
Polyline	6" CTS	Pipe Runs	None	93%
Pipe Tag	CTS 6"	Labels	None	93%
Polyline	6" CWS	Pipe Runs	None	96%
Pipe Tag	CWS 6"	Labels	None	96%
Polyline	6" CWR	Pipe Runs	None	94%
Pipe Tag	CWR 6"	Labels	None	94%
Gate valve	V-CWS-1	Valves & Devices	None	92%
Gate valve	V-CWR-1	Valves & Devices	None	91%
Butterfly valve	V-CTR-1	Valves & Devices	None	88%
Thermometer	TI-1	Valves & Devices	None	86%
Thermometer	TI-2	Valves & Devices	None	87%
Thermometer	TI-3	Valves & Devices	None	89%
Thermometer	TI-4	Valves & Devices	None	84%
Ball valve (drain/vent)	3/4" Ball valve (drain/vent)	Valves & Devices	None	79%
Ball valve (drain/vent)	3/4" Ball valve (drain/vent)	Valves & Devices	None	77%
Flange pair	6" Flange pair	Fittings	None	93%
Flange pair	6" Flange pair	Fittings	None	93%
45° elbow	6" 45° elbow	Fittings	None	90%
45° elbow	6" 45° elbow	Fittings	None	88%
90° LR elbow	6" 90° LR elbow	Fittings	None	71%
90° LR elbow	6" 90° LR elbow	Fittings	None	70%
Plate & frame heat exchanger	PHE-1	Valves & Devices	None	90%
Dimension	1'-10" C-C	Dimensions	None	82%
Dimension	11 3/4" C-C	Dimensions	None	80%
Dimension	3'-2"	Dimensions	None	74%
Stamp	AS-BUILT — Draft	Labels	None	100%
Cloud+	No support visible at CWR offset	Issues	None	72%
Cloud+	Nameplate not legible	Issues	None	90%

GENERAL NOTES

1. ALL PIPING 6" SCH 40 CS, 150# RF FLANGED U.N.O.
2. ELEVATIONS ESTIMATED FROM PHOTO — VERIFY IN FIELD
3. STENCILS READ: CTS · CTR · CWS · CWR

LEGEND

- CTR — Cooling Tower Return
- CTS — Cooling Tower Supply
- CWS — Condenser Water Supply
- CWR — Condenser Water Return
- Below 70% — verify

REVISION CLOUDS

△1 No support visible at CWR offset (Medium)
△2 Nameplate not legible (Low)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

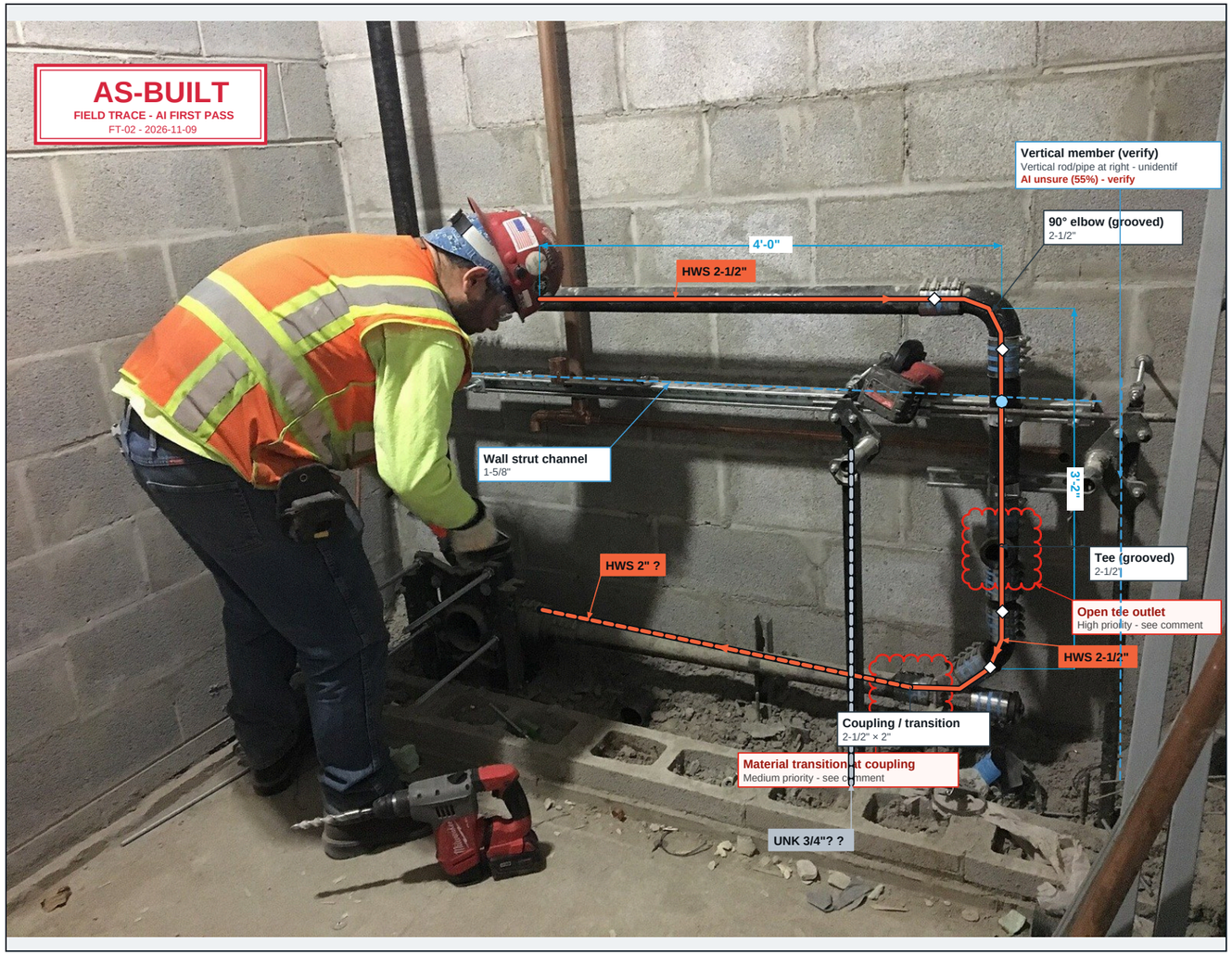
PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED FOR: Schilke Construction	
DRAWING TITLE: AS-BUILT — PLATE HX-1 — TOWER / CONDENSER WATER			LOCATION: Mech. Rm B-104 · North wall	
SOURCE PHOTO: FT-01 · 2026-11-09 07:42	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-01	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A SCHILKE JOB PHOTO · NOT FOR CONSTRUCTION

FIELD TRACE

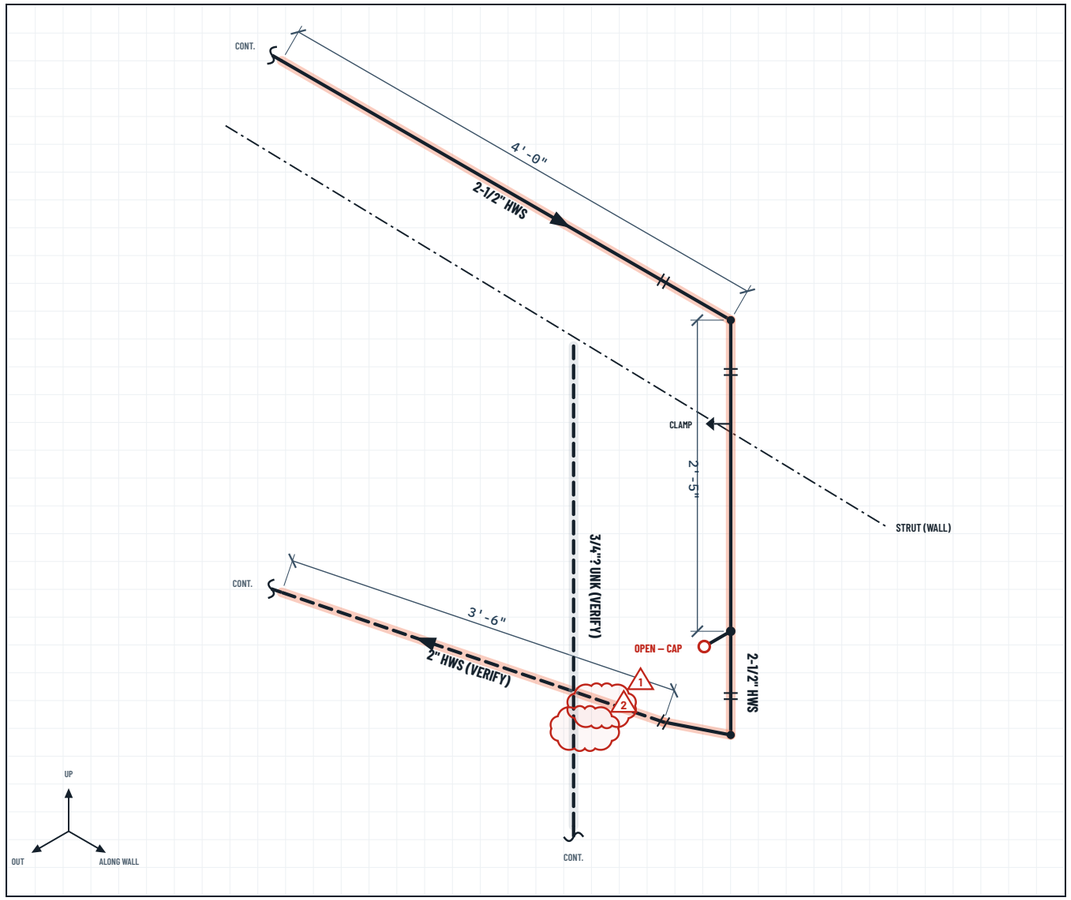
ANNOTATED FIELD PHOTO + AS-BUILT ISOMETRIC · FT-02

MIMETIC TECHNOLOGY PROTOTYPE · PREPARED FOR SCHILKE CONSTRUCTION · **SAMPLE DATA — NOT FOR CONSTRUCTION**



FIELD PHOTO FT-02 · LEVEL B1 · CORRIDOR C, COLUMN LINE 4 · 2026-11-09 09:15

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — HWS OFFSET AT CMU WALL — ROUGH-IN

N.T.S. · DERIVED FROM PHOTO FT-02 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 23 MARKUPS · 2 ISSUES · 6 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	2-1/2" HWS	Pipe Runs	None	91%
Pipe Tag	HWS 2-1/2"	Labels	None	91%
Polyline	2-1/2" HWS	Pipe Runs	None	86%
Pipe Tag	HWS 2-1/2"	Labels	None	86%
Polyline	2" HWS	Pipe Runs	None	68%
Pipe Tag	HWS 2"	Labels	None	68%
Polyline	3/4"? UNK	Pipe Runs	None	58%
Pipe Tag	UNK 3/4"?	Labels	None	58%
Rigid coupling	2-1/2" Rigid coupling	Fittings	None	93%
90° elbow (grooved)	2-1/2" 90° elbow (grooved)	Fittings	None	94%
Rigid coupling	2-1/2" Rigid coupling	Fittings	None	90%
Tee (grooved)	2-1/2" Tee (grooved)	Fittings	None	87%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Rigid coupling	2-1/2" Rigid coupling	Fittings	None	88%
90° elbow (grooved)	2-1/2" 90° elbow (grooved)	Fittings	None	83%
Coupling / transition	2-1/2" × 2" Coupling / transition	Fittings	None	70%
Wall strut channel	1-5/8" Wall strut channel	Hangers & Supports	None	92%
Pipe clamp at strut	2-1/2" Pipe clamp at strut	Hangers & Supports	None	80%
Vertical member (verify)	Vertical member (verify)	Hangers & Supports	None	55%
Dimension	4'-0"	Dimensions	None	81%
Dimension	3'-2"	Dimensions	None	78%
Stamp	AS-BUILT — Draft	Labels	None	100%
Cloud+	**Open tee outlet**	**Issues**	**None**	**85%**
Cloud+	**Material transition at coupling**	**Issues**	**None**	**68%**

GENERAL NOTES

1. 2-1/2" GROOVED STEEL, RIGID COUPLINGS U.N.O.
2. DIMENSIONS FROM CMU COURSING — ±2"
3. DASHED = BELOW CONFIDENCE THRESHOLD, VERIFY

LEGEND

- HWS — Heating Hot Water Supply
- UNK — Unidentified — verify
- Below 70% — verify

REVISION CLOUDS

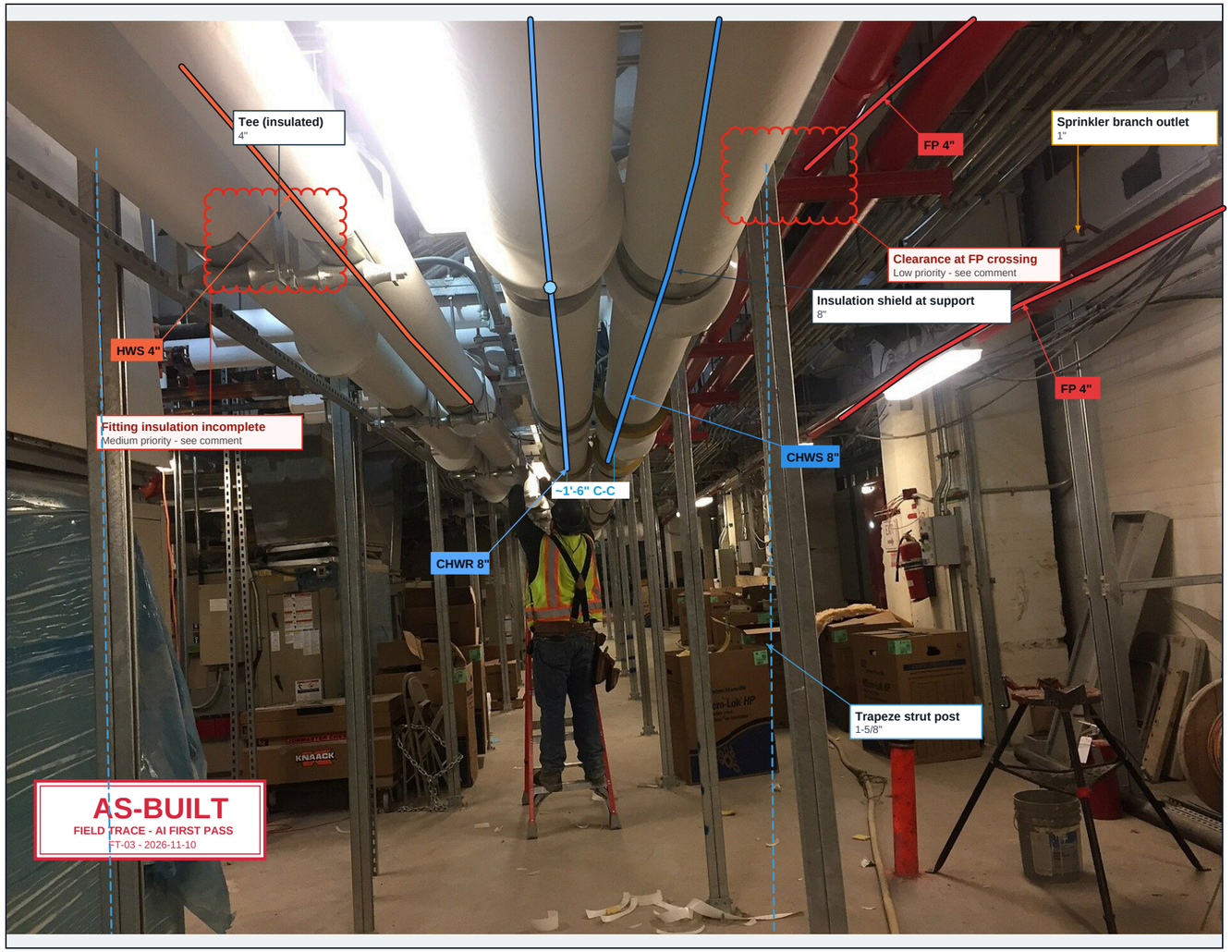
△1 Open tee outlet (High)
△2 Material transition at coupling (Medium)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

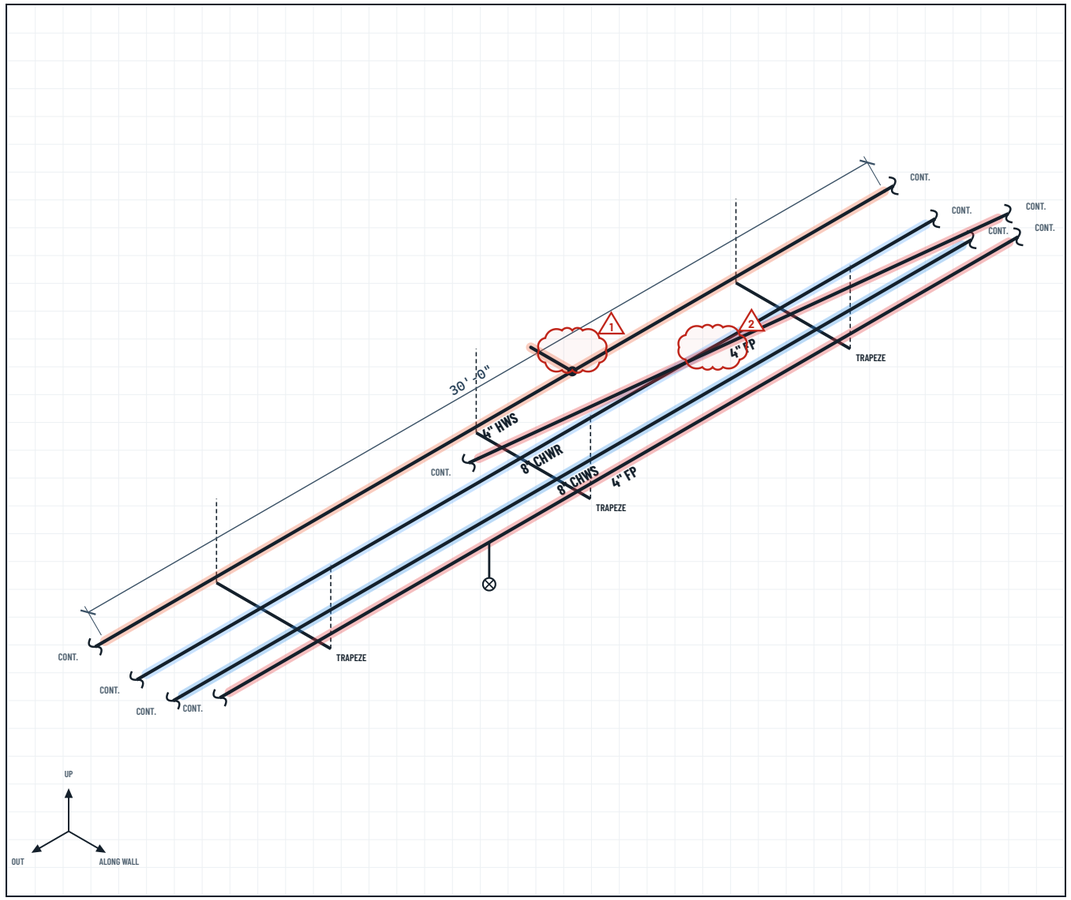
PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED FOR: Schilke Construction	
DRAWING TITLE: AS-BUILT — HWS OFFSET AT CMU WALL — ROUGH-IN			LOCATION: Level B1 · Corridor C, column line 4	
SOURCE PHOTO: FT-02 · 2026-11-09 09:15	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-02	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A SCHILKE JOB PHOTO · NOT FOR CONSTRUCTION



FIELD PHOTO FT-03 · LEVEL B1 · EAST SERVICE CORRIDOR · 2026-11-10 13:05

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — OVERHEAD MAINS — EAST SERVICE CORRIDOR

N.T.S. · DERIVED FROM PHOTO FT-03 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 20 MARKUPS · 2 ISSUES · 2 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	8" CHWS	Pipe Runs	None	88%
Pipe Tag	CHWS 8"	Labels	None	88%
Polyline	8" CHWR	Pipe Runs	None	86%
Pipe Tag	CHWR 8"	Labels	None	86%
Polyline	4" HWS	Pipe Runs	None	79%
Pipe Tag	HWS 4"	Labels	None	79%
Polyline	4" FP	Pipe Runs	None	90%
Pipe Tag	FP 4"	Labels	None	90%
Polyline	4" FP	Pipe Runs	None	84%
Pipe Tag	FP 4"	Labels	None	84%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Insulation shield at support	8" Insulation shield at support	Hangers & Supports	None	83%
Insulation shield at support	8" Insulation shield at support	Hangers & Supports	None	80%
Trapeze strut post	1-5/8" Trapeze strut post	Hangers & Supports	None	86%
Trapeze strut post	1-5/8" Trapeze strut post	Hangers & Supports	None	84%
Sprinkler branch outlet	1" Sprinkler branch outlet	Valves & Devices	None	74%
Tee (insulated)	4" Tee (insulated)	Fittings	None	70%
Dimension	~1'-6" C-C	Dimensions	None	62%
Stamp	AS-BUILT — Draft	Labels	None	100%
Cloud+	Fitting insulation incomplete	Issues	None	71%
Cloud+	Clearance at FP crossing	Issues	None	66%

GENERAL NOTES

1. RUN LENGTHS FROM ASSUMED 10'-0" SUPPORT SPACING
2. INSULATION: CHW 2", HW 1-1/2" (PER SPEC — VERIFY)
3. FP SHOWN FOR COORDINATION ONLY

LEGEND

- CHWS — Chilled Water Supply
- CHWR — Chilled Water Return
- HWS — Heating Hot Water Supply
- FP — Fire Protection
- Below 70% — verify

REVISION CLOUDS

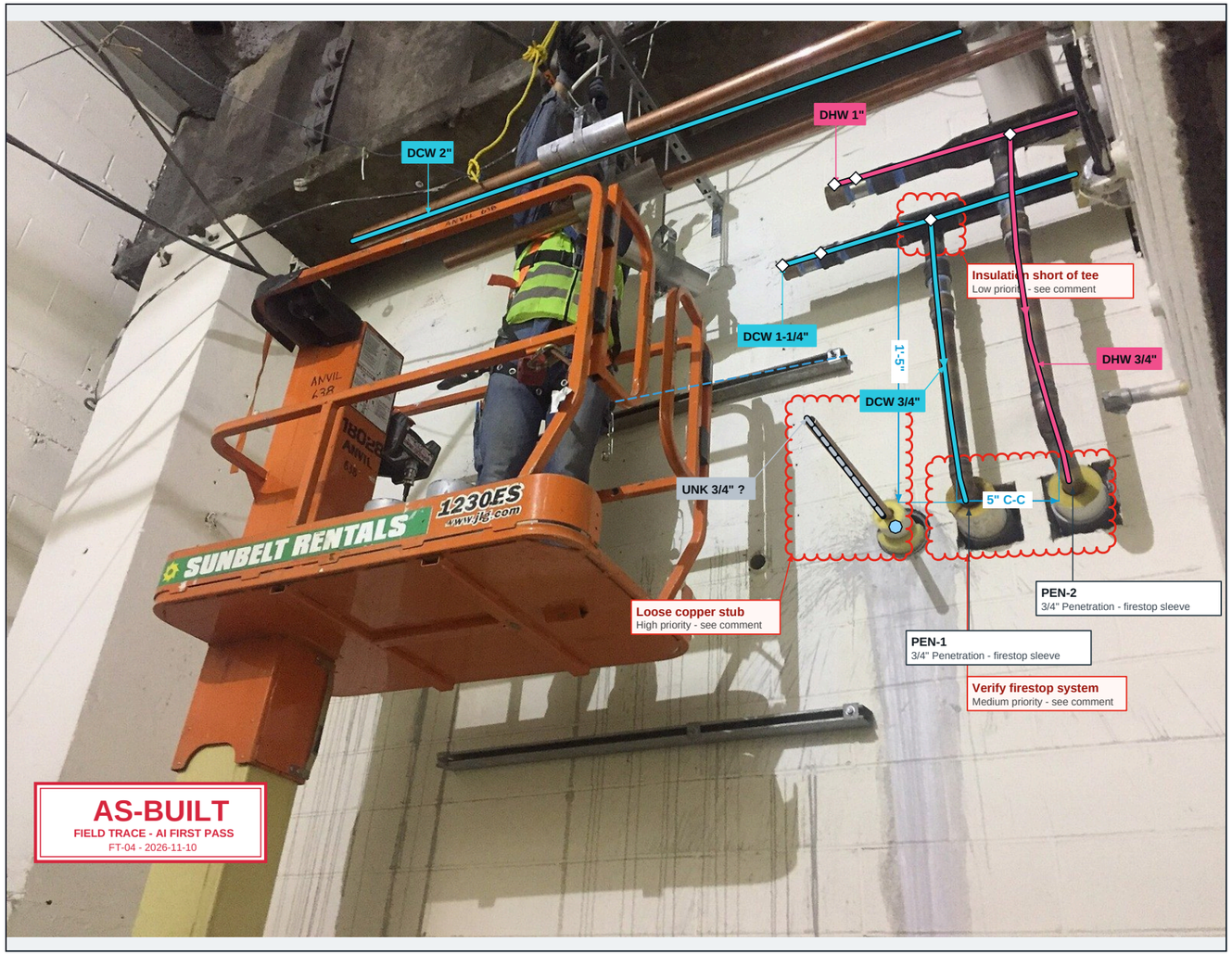
△1 Fitting insulation incomplete (Medium)
△2 Clearance at FP crossing (Low)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED FOR: Schilke Construction	
DRAWING TITLE: AS-BUILT — OVERHEAD MAINS — EAST SERVICE CORRIDOR			LOCATION: Level B1 · East service corridor	
SOURCE PHOTO: FT-03 · 2026-11-10 13:05	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-03	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A SCHILKE JOB PHOTO · NOT FOR CONSTRUCTION



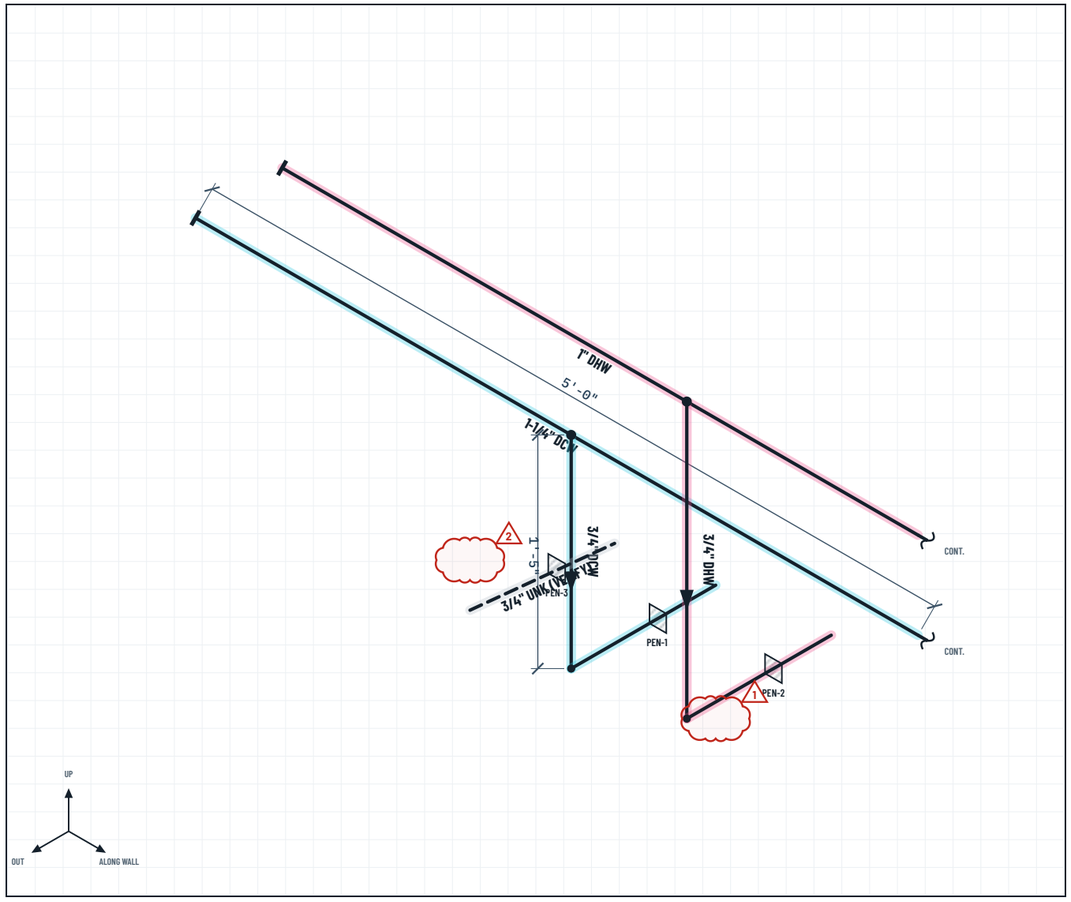
FIELD PHOTO FT-04 · LEVEL 1 · BACK-OF-HOUSE WALL W-12 · 2026-11-10 15:31

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI

MARKUP SUMMARY — 28 MARKUPS · 3 ISSUES · 4 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	1-1/4" DCW	Pipe Runs	None	85%
Pipe Tag	DCW 1-1/4"	Labels	None	85%
Polyline	1" DHW	Pipe Runs	None	83%
Pipe Tag	DHW 1"	Labels	None	83%
Polyline	3/4" DCW	Pipe Runs	None	88%
Pipe Tag	DCW 3/4"	Labels	None	88%
Polyline	3/4" DHW	Pipe Runs	None	76%
Pipe Tag	DHW 3/4"	Labels	None	76%
Polyline	2" DCW	Pipe Runs	None	72%
Pipe Tag	DCW 2"	Labels	None	72%
Polyline	3/4" UNK	Pipe Runs	None	60%
Pipe Tag	UNK 3/4"	Labels	None	60%
Tee	1-1/4" × 3/4" Tee	Fittings	None	84%
Tee (assumed)	1" × 3/4" Tee (assumed)	Fittings	None	66%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Press coupling	1" Press coupling	Fittings	None	82%
Press coupling	1-1/4" Press coupling	Fittings	None	82%
Cap (stub-out)	1" Cap (stub-out)	Fittings	None	75%
Cap (stub-out)	1-1/4" Cap (stub-out)	Fittings	None	75%
Penetration — firestop sleeve	PEN-1	Hangers & Supports	None	90%
Penetration — firestop sleeve	PEN-2	Hangers & Supports	None	90%
Penetration — firestop sleeve	PEN-3	Hangers & Supports	None	86%
Wall strut channel	1-5/8" Wall strut channel	Hangers & Supports	None	85%
Dimension	5" C-C	Dimensions	None	72%
Dimension	1'-5"	Dimensions	None	70%
Stamp	AS-BUILT — Draft	Labels	None	100%
Cloud+	Verify firestop system	Issues	None	80%
Cloud+	Loose copper stub	Issues	None	74%
Cloud+	Insulation short of tee	Issues	None	69%



AS-BUILT ISOMETRIC — DOMESTIC WATER DROPS AT WALL PENETRATIONS

N.T.S. · DERIVED FROM PHOTO FT-04 · DIMENSIONS ESTIMATED

GENERAL NOTES

1. TYPE L COPPER, PRESS FITTINGS U.N.O.
2. PENETRATIONS: SLEEVED, FIRESTOPPED — SYSTEM NO. TBD
3. R5 OVERHEAD 2" DCW NOT SHOWN (OCCLUDED)

LEGEND

- DCW — Domestic Cold Water
- DHW — Domestic Hot Water
- UNK — Unidentified — verify
- Below 70% — verify

REVISION CLOUDS

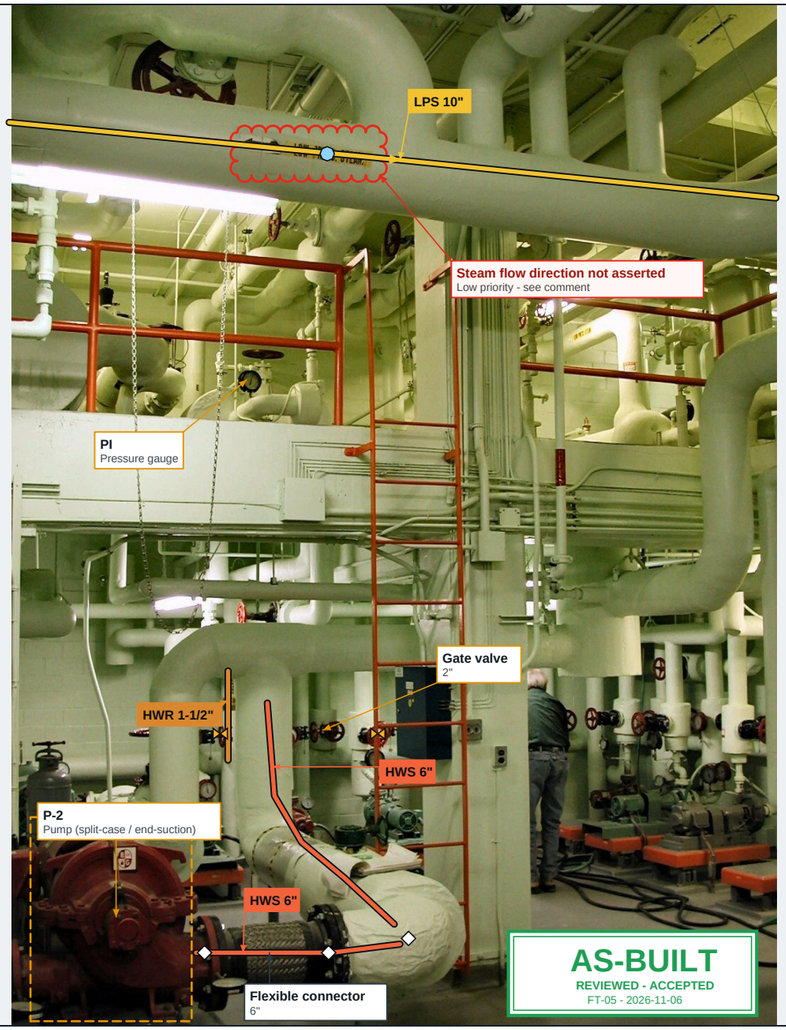
△1 Verify firestop system (Medium)
△2 Loose copper stub (High)
△3 Insulation short of tee (Low)

FIELD TRACE

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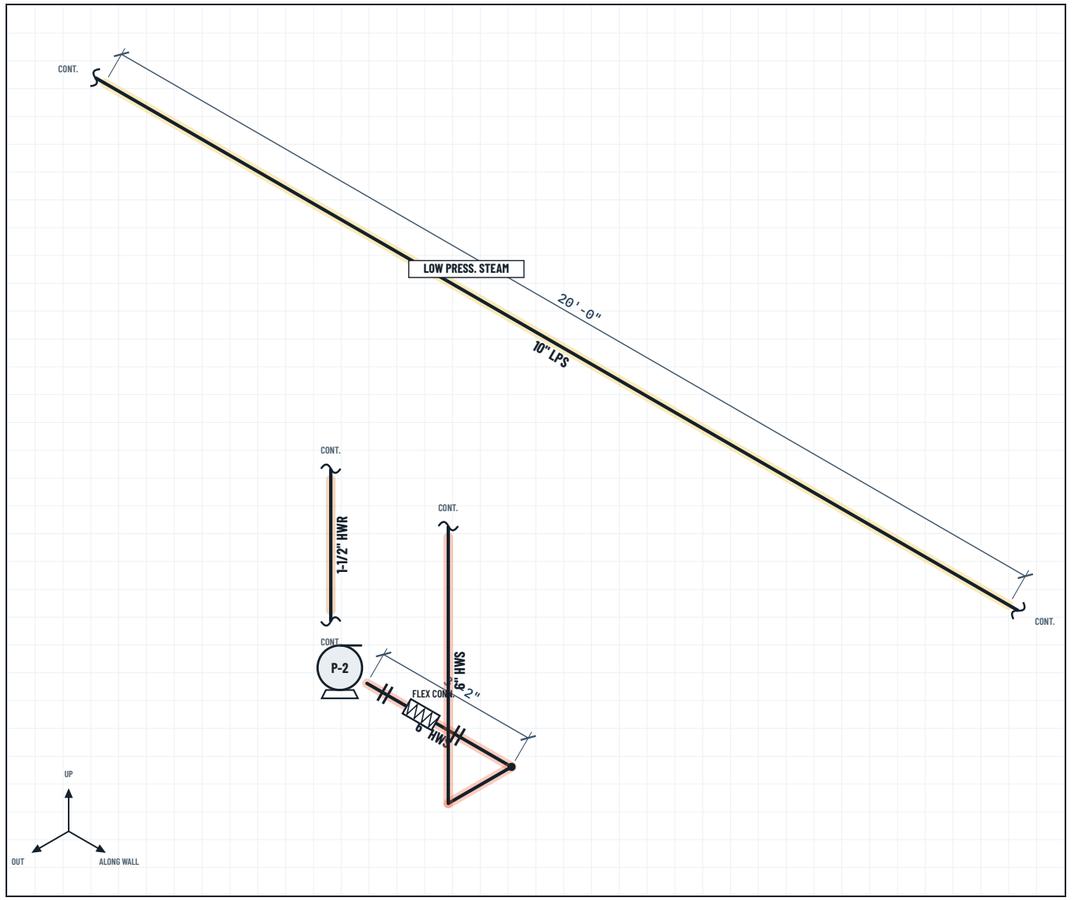
PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED FOR: Schilke Construction	
DRAWING TITLE: AS-BUILT — DOMESTIC WATER DROPS AT WALL PENETRATIONS			LOCATION: Level 1 · Back-of-house wall W-12	
SOURCE PHOTO: FT-04 · 2026-11-10 15:31	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-04	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A SCHILKE JOB PHOTO · NOT FOR CONSTRUCTION



FIELD PHOTO FT-05 · CENTRAL PLANT · PUMP ROOM · 2026-11-06 10:20

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — LP STEAM HEADER & PUMP SET

N.T.S. · DERIVED FROM PHOTO FT-05 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 20 MARKUPS · 1 ISSUES · 0 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	10" LPS	Pipe Runs	Accepted	93%
Pipe Tag	LPS 10"	Labels	Accepted	93%
Polyline	6" HWS	Pipe Runs	Accepted	87%
Pipe Tag	HWS 6"	Labels	Accepted	87%
Polyline	6" HWS	Pipe Runs	Accepted	74%
Pipe Tag	HWS 6"	Labels	Accepted	74%
Polyline	1-1/2" HWR	Pipe Runs	Accepted	70%
Pipe Tag	HWR 1-1/2"	Labels	Accepted	70%
Pump (split-case / end-suction)	P-2	Valves & Devices	Accepted	88%
Flexible connector	6" Flexible connector	Fittings	Accepted	92%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Flange pair	6" Flange pair	Fittings	Accepted	90%
Flange pair	6" Flange pair	Fittings	Accepted	90%
90° elbow (insulated)	6" 90° elbow (insulated)	Fittings	Accepted	82%
Gate valve	2" Gate valve	Valves & Devices	Accepted	78%
Gate valve	2" Gate valve	Valves & Devices	Accepted	78%
Gate valve	2" Gate valve	Valves & Devices	Accepted	76%
Pressure gauge	PI	Valves & Devices	Accepted	83%
Pipe marker (read)	LOW PRESS. STEAM	Labels	Accepted	95%
Stamp	AS-BUILT — Reviewed	Labels	Accepted	100%
Cloud+	Steam flow direction not asserted	Issues	Accepted	90%

GENERAL NOTES

1. REVIEWED SET — ALL MARKUPS ACCEPTED
2. CONTINUATIONS SHOWN WITH BREAK SYMBOL
3. HEADER LENGTH IN FRAME ≈ 20'-0" (ESTIMATED)

LEGEND

- LPS — Low Pressure Steam
- HWS — Heating Hot Water Supply
- HWR — Heating Hot Water Return
- Below 70% — verify

REVISION CLOUDS

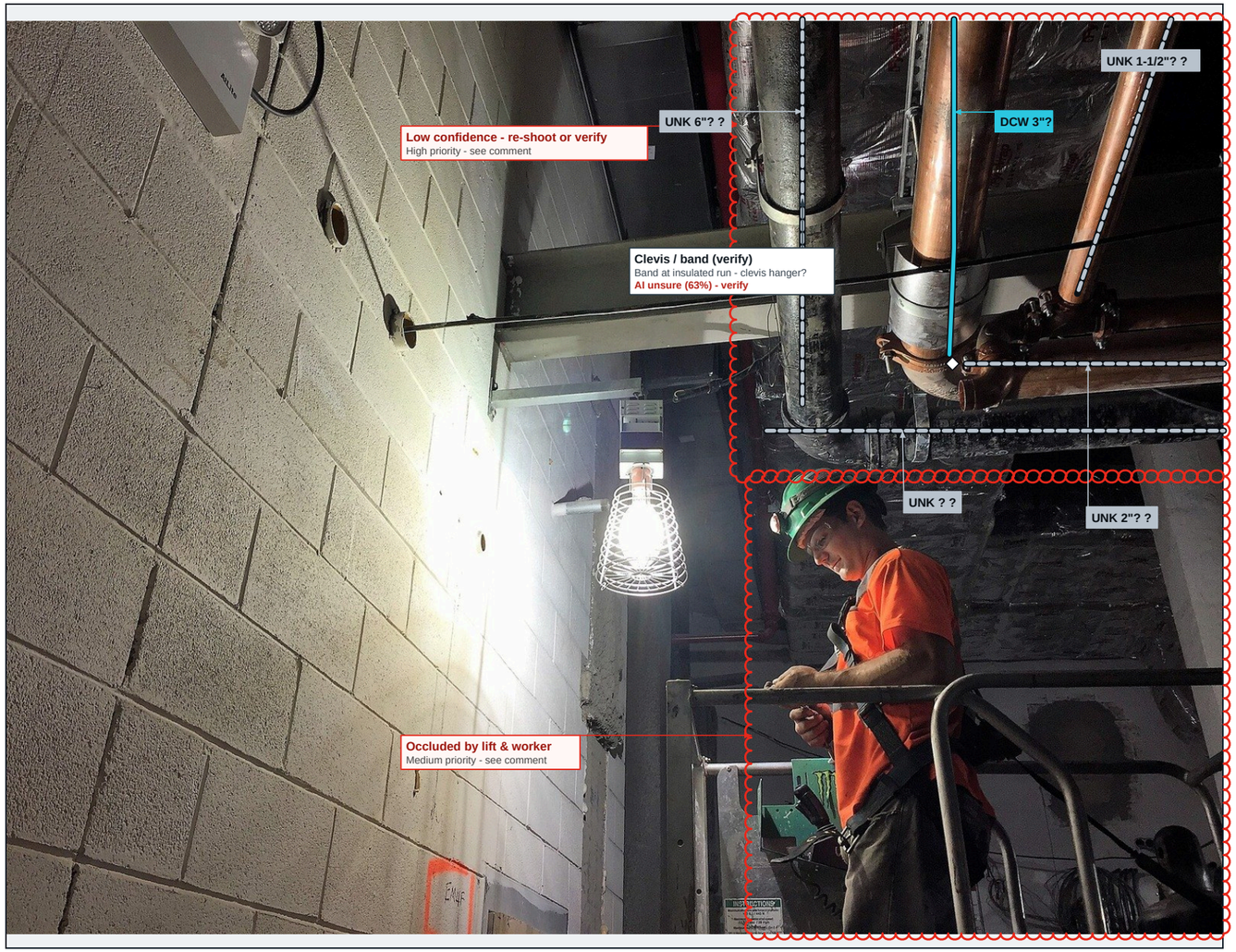
△1 Steam flow direction not asserted (Low)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

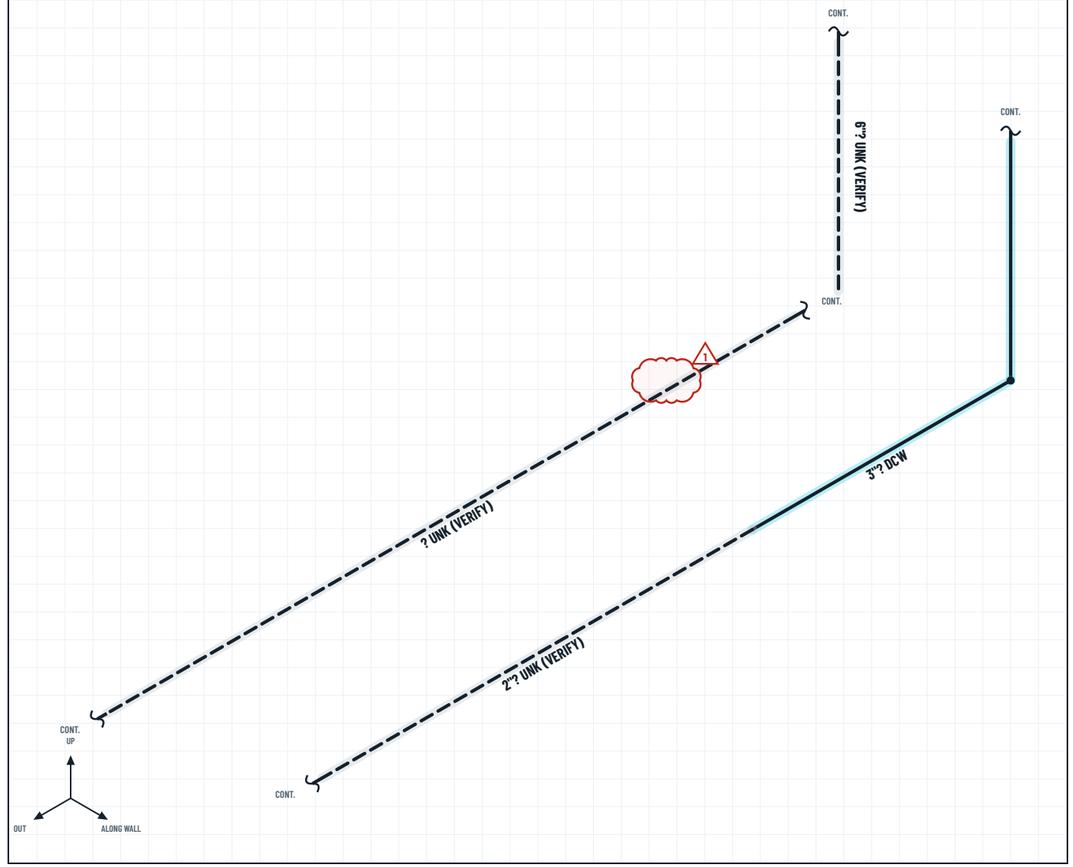
PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED FOR: Schilke Construction	
DRAWING TITLE: AS-BUILT — LP STEAM HEADER & PUMP SET			LOCATION: Central plant · Pump room	
SOURCE PHOTO: FT-05 · 2026-11-06 10:20	DRAWN: FIELD TRACE AI	CHECKED: R. (REVIEWER)	SCALE: N.T.S.	REV: 0
STATUS: REVIEWED — ISSUED AS-BUILT			SHEET: FT-05	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A SCHILKE JOB PHOTO · NOT FOR CONSTRUCTION



FIELD PHOTO FT-06 · LEVEL B1 · CORRIDOR D CEILING · 2026-11-11 16:48

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — CEILING SPACE ABOVE CORRIDOR — LOW LIGHT

N.T.S. · DERIVED FROM PHOTO FT-06 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 14 MARKUPS · 2 ISSUES · 10 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	6"? UNK	Pipe Runs	None	66%
Pipe Tag	UNK 6"?	Labels	None	66%
Polyline	3"? DCW	Pipe Runs	None	71%
Pipe Tag	DCW 3"?	Labels	None	71%
Polyline	2"? UNK	Pipe Runs	None	58%
Pipe Tag	UNK 2"?	Labels	None	58%
Polyline	1-1/2"? UNK	Pipe Runs	None	61%
Pipe Tag	UNK 1-1/2"?	Labels	None	61%
Polyline	? UNK	Pipe Runs	None	52%
Pipe Tag	UNK ?	Labels	None	52%
Clevis / band (verify)	Clevis / band (verify)	Hangers & Supports	None	63%
Elbow (copper)	3"? Elbow (copper)	Fittings	None	60%
Cloud+	Low confidence — re-shoot or verify	Issues	None	95%
Cloud+	Occluded by lift & worker	Issues	None	90%

GENERAL NOTES

1. NOT ISSUED — HELD FOR REVIEW
2. DASHED RUNS BELOW 70% CONFIDENCE
3. SIZES UNVERIFIED (NO SCALE REFERENCE)

LEGEND

- UNK — Unidentified — verify
- DCW — Domestic Cold Water
- Below 70% — verify

REVISION CLOUDS

△1 Low confidence — re-shoot or verify (High)

△2 Occluded by lift & worker (Medium)

FIELD TRACE

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80/20 SOFTWARE · PROTOTYPE

PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED FOR: Schilke Construction	
DRAWING TITLE: AS-BUILT — CEILING SPACE ABOVE CORRIDOR — LOW LIGHT			LOCATION: Level B1 · Corridor D ceiling	
SOURCE PHOTO: FT-06 · 2026-11-11 16:48	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: HELD — NEEDS FIELD VERIFICATION			SHEET: FT-06	

PIPE — BY SYSTEM / SIZE / MATERIAL

SYSTEM	SIZE	MATERIAL	LF	PHOTOS	
CHWR Chilled Water Return	8"	CS, 2" insulation w/ PVC jacket	30.0	FT-03	
CHWS Chilled Water Supply	8"	CS, 2" insulation w/ PVC jacket	30.0	FT-03	
CTR Cooling Tower Return	6"	Sch 40 CS, flanged	5.9	FT-01	
CTS Cooling Tower Supply	6"	Sch 40 CS, flanged	9.9	FT-01	
CWR Condenser Water Return	6"	Sch 40 CS, flanged	5.9	FT-01	
CWS Condenser Water Supply	6"	Sch 40 CS, flanged	9.6	FT-01	
DCW Domestic Cold Water	3"?	Copper	5.4	FT-06	
DCW Domestic Cold Water	1-1/4"	Type L copper, insulated	5.0	FT-04	
DCW Domestic Cold Water	3/4"	Type L copper	2.4	FT-04	
DCW Domestic Cold Water	2"	Type L copper	0.0	FT-04	
DHW Domestic Hot Water	1"	Type L copper, insulated	4.4	FT-04	
DHW Domestic Hot Water	3/4"	Type L copper	2.9	FT-04	
FP Fire Protection	4"	Sch 10 CS, red	48.1	FT-03	
HWR Heating Hot Water Return	1-1/2"	CS, insulated	2.5	FT-05	
HWS Heating Hot Water Supply	4"	CS, 1-1/2" insulation	31.6	FT-03	
HWS Heating Hot Water Supply	2-1/2"	Sch 10 CS, grooved	7.8	FT-02	
HWS Heating Hot Water Supply	6"	CS, insulated	5.1	FT-05	
HWS Heating Hot Water Supply	6"	CS	4.6	FT-05	
HWS Heating Hot Water Supply	2"	Galv.? — verify	3.5	FT-02	VERIFY
LPS Low Pressure Steam	10"	CS, insulated	20.0	FT-05	
UNK Unidentified — verify	?	Insulated (black)	8.0	FT-06	VERIFY
UNK Unidentified — verify	2"?	Copper	5.0	FT-06	VERIFY
UNK Unidentified — verify	3/4"?	Unknown	3.7	FT-02	VERIFY
UNK Unidentified — verify	6"?	Insulated, metal jacket	2.5	FT-06	VERIFY
UNK Unidentified — verify	3/4"	Copper	1.0	FT-04	VERIFY
UNK Unidentified — verify	1-1/2"?	Copper	0.0	FT-06	VERIFY
Total			254.9		

OPEN ISSUES

PHOTO	SEVERITY	ISSUE	COMMENT
FT-01	Medium	**No support visible at CWR offset**	6" CWR 45° offset between valve and HX shows no hanger or floor stand in frame. Verify support per spec before insulation/close-out.
FT-01	Low	**Nameplate not legible**	PHE-1 tag assumed from drawing schedule. Capture nameplate close-up (model / serial) for O&M.
FT-02	High	**Open tee outlet**	Branch outlet on 2-1/2" tee is open. Cap or connect before hydrostatic test.
FT-02	Medium	**Material transition at coupling**	Finish changes at coupling (black → galvanized?). If dissimilar metals, verify dielectric/transition fitting per spec.
FT-03	Medium	**Fitting insulation incomplete**	Insulation at tee cluster is irregular / covers appear missing. Complete fitting covers before ceiling close-in.
FT-03	Low	**Clearance at FP crossing**	Upper FP main crosses tight above CHWS. Verify clearance and access to FP fittings.
FT-04	Medium	**Verify firestop system**	Sealant present at 3 sleeves. Confirm listed firestop system number and label each penetration per spec.

FITTINGS, VALVES & SUPPORTS

CATEGORY	ITEM	SIZE	QTY	PHOTOS	
Fittings	45° elbow	6"	2	FT-01	
Fittings	90° elbow (grooved)	2-1/2"	2	FT-02	
Fittings	90° elbow (insulated)	6"	1	FT-05	
Fittings	90° LR elbow	6"	2	FT-01	
Fittings	Cap (stub-out)	1"	1	FT-04	
Fittings	Cap (stub-out)	1-1/4"	1	FT-04	
Fittings	Coupling / transition	2-1/2" × 2"	1	FT-02	
Fittings	Elbow (copper)	3"?	1	FT-06	VERIFY
Fittings	Flange pair	6"	4	FT-01 FT-05	
Fittings	Flexible connector	6"	1	FT-05	
Fittings	Press coupling	1"	1	FT-04	
Fittings	Press coupling	1-1/4"	1	FT-04	
Fittings	Rigid coupling	2-1/2"	3	FT-02	
Fittings	Tee	1-1/4" × 3/4"	1	FT-04	
Fittings	Tee (assumed)	1" × 3/4"	1	FT-04	VERIFY
Fittings	Tee (grooved)	2-1/2"	1	FT-02	
Fittings	Tee (insulated)	4"	1	FT-03	
Hangers, supports & penetrations	Clevis / band (verify)	—	1	FT-06	VERIFY
Hangers, supports & penetrations	Insulation shield at support	8"	2	FT-03	
Hangers, supports & penetrations	Penetration — firestop sleeve	3/4"	3	FT-04	
Hangers, supports & penetrations	Pipe clamp at strut	2-1/2"	1	FT-02	
Hangers, supports & penetrations	Trapeze strut post	1-5/8"	2	FT-03	
Hangers, supports & penetrations	Vertical member (verify)	—	1	FT-02	VERIFY
Hangers, supports & penetrations	Wall strut channel	1-5/8"	2	FT-02 FT-04	
Valves & devices	Ball valve (drain/vent)	3/4"	2	FT-01	
Valves & devices	Butterfly valve	6"	1	FT-01	
Valves & devices	Gate valve	6"	2	FT-01	
Valves & devices	Gate valve	2"	3	FT-05	
Valves & devices	Plate & frame heat exchanger	—	1	FT-01	
Valves & devices	Pressure gauge	—	1	FT-05	
Valves & devices	Pump (split-case / end-suction)	—	1	FT-05	
Valves & devices	Sprinkler branch outlet	1"	1	FT-03	
Valves & devices	Thermometer	—	4	FT-01	

Lengths are taken from the as-built model drafted from each photo, with estimated dimensions. Rejected markups are left out. Items under 70% confidence are counted and flagged. Not for bidding. This shows how the takeoff would roll up.